THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** acidic	Acid turns blue litmus red and red remains red.
2	**(c)** 6.0 V	4 x 1.5.
3	**(d)** metal	Metal properties.
4	**(a)** burning of coal, ripening of fruit	Both produce new substances.
5	**(c)** NaOH	NaOH.
6	**(a)** using damaged wires	Damaged wires are dangerous.
7	**(d)** red litmus blue	Basic.
8	**(a)** no moisture	Moisture is needed.
9	**(a)** it contains acetic acid	Acid.
10	**(d)** becomes brighter	More voltage.
11	**(c)** a protective oxide layer forms	Al2O3 layer.
12	**(c)** crystals form when a hot saturated solution cools	Cooling.
13	**(a)** basic	Typical of bases.
14	**(a)** 0 V	They cancel.
15	**(b)** non-metal	Non-metal.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	A base (1); Soap solution (1).	1+1
17	Coin is a conductor, completing the circuit (1); Plastic is an insulator (1).	1+1
18	Silver reacts with sulphur compounds in air (1); Forms black silver sulphide (1).	1+1
19	Melting of wax is physical (1); Burning of wax vapour is chemical (1).	1+1
20	$\mathrm{H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O}$ (1); Salt: sodium sulphate (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	X: basic (1); Y: acidic (1); Reason from colours (1).	1+1+1
22	Total voltage 6 V (1); Matches bulb rating (1); Bulb glows normally (1).	1+1+1
23	No reaction (1); Copper is less reactive (1); It cannot displace iron (1).	1+1+1
24	Nail turns brown (1); Solution turns pale green (1); Chemical change (1).	1+1+1
	OR: Both contain bases (1); Neutralise stomach acid (1); Rate of action differs (1).	1+1+1
25	Senses large current (1); Switch trips (1); Can be reset (1).	1+1+1
	OR: Coating is basic copper carbonate or oxide (1); Acid in lemon reacts (1); Surface becomes shiny (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Test with blue and red litmus (1); Two turn blue litmus red: vinegar, lemon (1); Two turn red litmus blue: lime water, soap (1); Sugar: no change (1); Use taste or smell safe methods (1).	1+1+1+1+1
	OR: Cell, bulb, wires, free ends (1); Place samples: iron nail, coin, eraser, plastic comb, graphite (1); Bulb glows for conductors (1); Table of results (1); Conclude metals and graphite conduct (1).	1+1+1+1+1
27	Collect samples (1); Observe lustre (1); Test conduction (1); Strike and listen (1); Table and conclusion (1).	1+1+1+1+1

	OR: Two beakers: tap water and salt water (1); Same nails and exposure (1); Observe for a week (1); Salt water nail rusts more (1); Salt speeds rusting (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** freezing of water, evaporation of water	States change only.
2	**(a)** sodium sulphate	Na_2SO_4.
3	**(b)** are insulators	Block current.
4	**(b)** blue litmus red	Acid.
5	**(c)** no air reaches it	Oil stops oxygen.
6	**(a)** it is a base	Base.
7	**(a)** a cell, bulb and wires	Must have a source and indicator.
8	**(d)** it is lustrous	Lustre.
9	**(d)** to remove insoluble impurities	Clear solution.
10	**(c)** neutral	Neutral.
11	**(a)** the filament is broken or the circuit is open	Gap in path.
12	**(a)** lead	Lead is a poor conductor among metals.
13	**(b)** frying of an egg	Chemical change.
14	**(d)** potassium nitrate	KNO_3.
15	**(c)** each bulb terminal connected to a different cell terminal	Complete path.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Voltages cancel (1); Bulb does not glow (1).	1+1
17	Magnesium hydroxide as antacid (1); Calcium oxide for neutralising acid (1).	1+1
18	Solubility decreases with temperature fall (1); Extra solute separates (1).	1+1
19	Salt is formed in neutralisation (1); Acid is sour and turns blue litmus red (1).	1+1
20	Switch off the main supply (1); Use the same rating (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Plants make glucose (1); Uses carbon dioxide and water (1); New substances formed (1).	1+1+1
22	Avoid confusion (1); Safety (1); Correct use (1).	1+1+1
23	Switch off (1); Hold the plug, not the wire (1); Dry hands (1).	1+1+1
24	Heat spreads (1); Food cooks quickly (1); Handles remain cool (1).	1+1+1
	OR: Cooking: chemical (1); Melting: physical (1); Irreversible vs reversible (1).	1+1+1
25	Boil pieces (1); Filter extract (1); Vinegar gives red, soap gives green (1).	1+1+1
	OR: Cell, bulb, wires and two free ends (1); Touch objects across ends (1); Conductors make the bulb glow (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Fizzing (1); Gas released (1); Carbon dioxide (1); Acid + base reaction (1); Lime water turns milky with the gas (1).	1+1+1+1+1
	OR: Switch off main supply (1); Use insulated tools (1); Wear rubber gloves (1); Check with a tester (1); Stand on dry insulated mat (1).	1+1+1+1+1
27	Hold with tongs (1); Do not look directly (1); Collect the ash (1); Dissolve in water (1); Test with litmus (1).	1+1+1+1+1
	OR: Useful: cooking (1); Useful: making curd (1); Useful: photosynthesis (1); Harmful: rusting (1); Harmful: food spoiling (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** milky	Calcium carbonate.
2	**(a)** both air and water are present	Conditions fulfilled.
3	**(d)** more lactic acid forms	Acid increases.
4	**(b)** glow	Graphite conducts.
5	**(b)** copper conducts, PVC insulates	Safety.
6	**(a)** slow cooling gives bigger crystals	Time for growth.
7	**(d)** soap solution	Basic.
8	**(c)** the switch is closed	Closed circuit.
9	**(d)** graphite	Carbon form.
10	**(a)** dissolving is a physical change	Original substance recovered.
11	**(b)** calcium chloride	$CaCl_2$.
12	**(d)** more cells are added in series	More voltage.
13	**(b)** hydrogen	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$.
14	**(c)** salt in sea air speeds rusting	Salt accelerates.
15	**(d)** to neutralise mouth acids	Prevents decay.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It catches fire in air (1); Water prevents contact (1).	1+1
17	Zinc coating protects (1); Zinc corrodes first (1).	1+1
18	Vinegar is an acid and neutralises the base (1); Baking soda is also a base so it cannot neutralise it (1).	1+1
19	Long wire fits in a small space (1); More heat in a small region (1).	1+1
20	It forms a protective oxide layer (1); The layer protects the metal inside (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Take three test tubes (1); Add metal strips (1); Zinc bubbles fast, iron slowly, copper not at all (1).	1+1+1
22	Fizzing (1); Carbon dioxide (1); Chemical change (1).	1+1+1
23	Metals react with acids (1); Make the food unsafe (1); Use glass or ceramic (1).	1+1+1
24	Geysers need more current (1); Thick wires carry it safely (1); Thin wires would overheat (1).	1+1+1
	OR: Iron: hard, magnetic, conducts (1); Sulphur: brittle, yellow, insulator (1); Different type (1).	1+1+1
25	Oxygen causes spoilage (1); Moisture causes spoilage (1); Keeps it fresh (1).	1+1+1
	OR: Vinegar (1); Lemon (1); Baking soda (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Add litmus to acid: red (1); Add base drop by drop (1); Colour turns violet (1); $HCl + NaOH \longrightarrow NaCl + H_2O$ (1); Evaporate to get salt (1).	1+1+1+1+1
	OR: Connect nichrome wire to cells (1); Pass current (1); Wire gets hot (1); Increase cells: hotter (1); Warn about burns (1).	1+1+1+1+1
27	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $S + O_2 \longrightarrow SO_2$ (1); Dissolve in water (1); MgO basic, SO2 acidic (1); Litmus test (1).	1+1+1+1+1
	OR: Fuel burns: chemical (1); Engine heats: physical (1); Paint protects (1); Rust forms: chemical (1); Exhaust gases (1).	1+1+1+1+1

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